

At last, a data-centric OS?



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The recent announcement by Google of the forthcoming Chrome OS immediately reminded me of a prediction I had heard in the mid-90s, and suggested that Google might actually be heading in the right direction.

In February 1995, I was invited to visit IBM's T. J. Watson Research Centre, just outside of New York. I had shared some obscure research data which happened to be of importance to IBM, and so this was a courtesy visit – and not a PR or marketing man in sight. One exercise the researchers do each year is to make predictions for the computer of ten years time. They also check each year to see what predictions they had made ten years before, and how accurate they were. (They proved very accurate at the time I was there, apart from the 40 MB floppy drive!)

I wrote at the time: “Almost sheepishly, they told me of one of their current predictions, which they seemed to feel confident about, but a little hesitant to express, apparently because it seemed so radical: that the user interface of the future would be Mosaic – or whatever was the successful browser at the time.” NCSA Mosaic was the first successful web browser, before it was eclipsed by Netscape.

Those researchers considered that the future importance of the web and the logic of having the same kind of interface to access local data and web-based data would force a demand for a browser style in the operating system interface.

For most of the time that we have had PCs, the interface has been program-centric. With command lines, such as DOS, you might enter: `edit column.txt`

The text file to be edited is a parameter for the command. You might also simply start the editor and then proceed to open various documents. Everything is done from the program. With a graphical interface like Windows, you also normally start a program first and from there open data files.

In a data-centric approach, typically, you might click on the icon of a text file and the OS will know which program should open the file – there are problems with this approach – click on a video clip, and do you want to view it, edit it or resample it?

There are other ways also in which the program-centric approach has advantages. Take my own work; I spend most of my time on my PCs

working with text: writing, reading or editing. I have three text editors that I use for this purpose, and having three icons on my desktop does not take up too much space. However, as I write, my main directory tree for documents contains over 5,000 text files – many very old ones are zipped up and not included in this count. Imagine that lot dropped on the desktop as icons.

However, the main attraction of developing a data-centric approach to the user interface lies in the fact that users think in a data-centric manner. I don't turn on my PC in the morning and think “Now I want to work with my word-processor.” Rather: “I need to get back to that document I was working on last night.”

So, how to develop a data-centric user interface? The idea has been around for a long time. Bill Gates told me around 1989 of his vision that he called “information at your fingertips”. He described this as an approach to all data for which the only analogy he could give at the time was hypertext (known at that time in program help systems). He had been very impressed with the work that had gone on at Xerox PARC, admitting in private that this was the inspiration for Windows, and that group was one of those working on hypertext. Perhaps some of you also remember the discussions in the 90s about the Program Manager for Windows and OS/2 – how to do this for data?

Gates was nearly right with the hypertext suggestion, but Microsoft seems never to have been able to embrace this fully. Clearly, a data-centric approach favours a GUI and browser-style interface. If anything has shown us how we can manage data on a large scale and find our way around it, it is the experience with web browsing. This is what the IBM researchers figured out nearly 15 years ago.

And this seems to be how the Google developers are thinking. Don't be fooled that this is only for netbooks – if they get it right, it will be made available for desktops. The irony is that just as Google is describing a browser-style interface for an operating system, Microsoft moves the other way and tells us that Internet Explorer is not an integral part of Windows and can be removed. That must be music to Google's ears...

A handwritten signature in dark ink, appearing to read "E. Henning". The signature is fluid and cursive, with a long, sweeping underline that extends to the right.

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